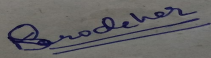


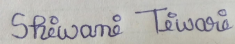
# AI-Powered Driver Drowsiness Detection System

## Milestone 1 Presentation

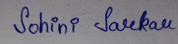
Team: Kushagra, Shiwani, Sohini, Shubham, Ravina



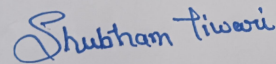
Kushagra



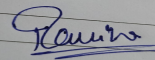
Shiwani Tiwari



Sohini Saxena



Shubham Tiwari

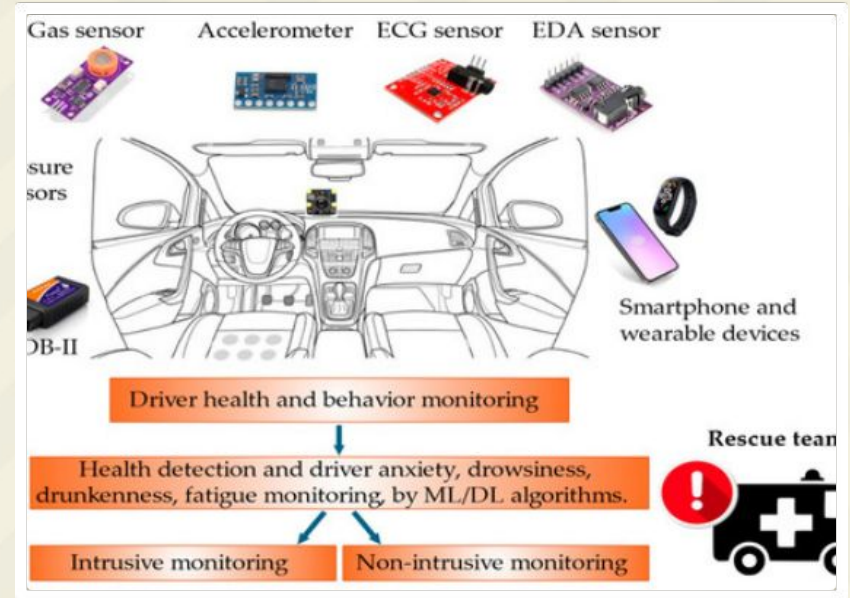


Ravina

# PROBLEM & MOTIVATION

## Safety Crisis on Roads

- **Night & Long-haul Safety:** Driver fatigue is a leading cause of fatal accidents in commercial transport.
- **Urbanization Trends:** Increasing cab and private vehicle usage demands automated alertness monitoring.
- **Proactive Warnings:** The system aims to provide an additional safety layer before fatigue becomes dangerous.



# SCOPE & STAKEHOLDERS

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## Hardware Scope

Prioritizing

**Dashboard/Embedded**

**Cameras** over smartphones to avoid battery drain and multitasking interference.



## Stakeholders

Logistics agencies, cab owners, bus operators, and individual drivers aiming for highway safety parameters.

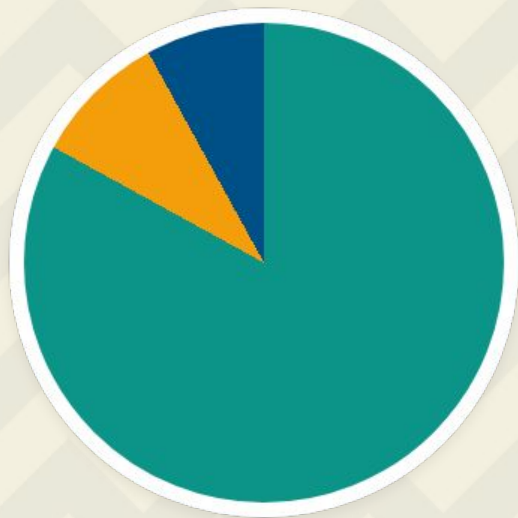


## Exclusions

Out of Scope: Real-car mechanical braking integration, mobile application development, and cloud dashboards.

# DATASET UNDERSTANDING

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## Roboflow Driver Dataset

**21,668 Total Labeled Images**

- Training Set: 17,961 (83%)
- Validation Set: 1,881 (9%)
- Testing Set: 1,826 (8%)

# CLASSES & HORIZON

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## Class Distribution

- **Open Eyes:** 9,510 objects
- **Yawn States:** 7,196 objects
- **Closed Eyes:** 6,804 objects

## Next Steps: Temporal Data

Integrating video frameworks for sequence validation in Milestone 2:

- **NTHU Dataset:** For sequence tracking and varied lighting profiles.
- **YawDD Dataset:** For specialized continuous yawning validation.



# RULE-BASED: MEDIAPIPE

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## Facial Landmark Logic

- **EAR (Eye Aspect Ratio):** Calculating distances between vertical eyelid points to isolate blinks and closures.
- **MAR (Mouth Aspect Ratio):** Monitoring vertical lip separation to identify distinct yawning trends.
- **OpenCV:** Real-time frame processing and handling runtime threshold overlays.



# PERCLOS & POSE LOGIC

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## PERCLOS

Tracking the percentage of eye closure over time to filter out standard, rapid blinks from genuine fatigue.



## Head Pose

Detecting nodding or slumping events by measuring head tilt (pitch/yaw) indicating fatigue.



## Limitations

Fixed ratios lose accuracy during lighting shifts, camera angle offsets, or when drivers wear eyewear.

# DEEP LEARNING: YOLO

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## Robust State Identification

- **Direct Classification:** Detecting Closed Eye or Yawn boxes directly from raw pixels.
- **Resiliency:** Convolutional weights adapt better to cabin noise, obstructions, and low lighting.
- **Models:** Utilizing **YOLOv8n** or **YOLO11n** for high FPS on dashboard hardware.



# PROPOSED HYBRID PIPELINE

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Input Feed

MediaPipe

YOLO Analysis

Sliding Window

Staged Alert

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**Proposed Paradigm:** Landmark precision cross-checked against YOLO detections, smoothed by temporal sliding windows for reliable, real-time alerts.

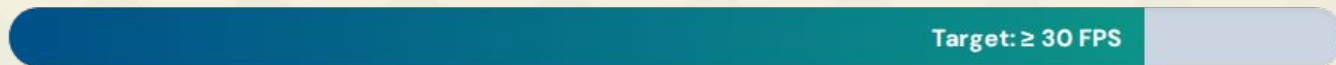
# EVALUATION METRICS

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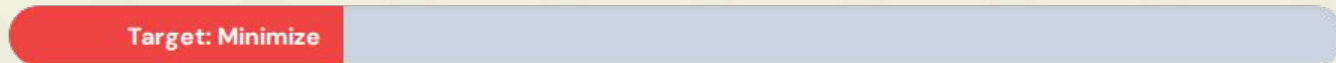
Precision & Recall



Processing Speed



False Alarm Rate



Testing via **mAP** layers and tracking stability over time using balanced **F1-scores**.

# TESTING & ALERTS

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## Environments

Validating across lighting challenges, tunnel simulations, eyewear variations, and deep angle changes.



## Alert Logic

**Stage 1:** Alert State (Green)

**Stage 2:** Mild Fatigue (Orange Visual)

**Stage 3:** Drowsy Warning (Red Audio)



## Expected Output

Continuous fatigue score log records paired with immediate onscreen tracking overlays.

# REFERENCES

Roboflow Driver Drowsiness Public Dataset Engine Hub (2024)

MediaPipe Face Mesh Processing Framework – Google Engineering

Ultralytics YOLO Framework Architecture Repository

Soukupova & Cech: Real-Time Eye Blink Detection (EAR Reference)

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# Thank You!

Questions? | Milestone 1 Final Presentation